Chapter 5, Problem 42P

Problem

The feedback resistor of a three-input averaging summing amplifier is 50 kΩ. What are the values of *R*1, *R*2, and *R*3?

Step-by-step solution

Step 1 of 4

The output equation of three input summing amplifier is,

$$v_0 = -R_f\left(\frac{v_1}{R_1}+\frac{v_2}{R_2}+\frac{v_3}{R_3}\right)$$

$$v_0 = -\left(\frac{R_f}{R_1}v_1+\frac{R_f}{R_2}v_2+\frac{R_f}{R_3}v_3\right) \quad \text{...... (1)}$$

An averaging amplifier is a summer that provides an output equal to the average of the inputs. The output equation of averaging amplifier with three inputs is,

$$v_0 = -\frac{1}{3}\left(v_1+v_2+v_3\right)$$

$$v_0 = -\left(\frac{1}{3}v_1+\frac{1}{3}v_2+\frac{1}{3}v_3\right) \quad \text{...... (2)}$$

Step 2 of 4

Compare first terms of equations (1) and (2) to get the value of R_1.

$$\frac{R_f}{R_1}=\frac{1}{3}$$

$$R_1 = 3R_f$$

Substitute $50\,\text{k}\Omega$ for R_f.

$$R_1 = 3\left(50\times10^3\right)$$

$$R_1 = 150\,\text{k}\Omega$$

Therefore, the value of R_1, is $\boxed{150\,\text{k}\Omega}$.

Step 3 of 4

Compare second terms of equations (1) and (2) to get the value of R_2.

$$\frac{R_f}{R_2}=\frac{1}{3}$$

$$R_2 = 3R_f$$

Substitute $50\,\text{k}\Omega$ for R_f.

$$R_2 = 3\left(50\times10^3\right)$$

$$R_2 = 150\,\text{k}\Omega$$

Therefore, the value of R_2, is $\boxed{150\,\text{k}\Omega}$.

Compare third terms of equations (1) and (2) to get the value of R_3.

$$\frac{R_f}{R_3} = \frac{1}{3}$$

$$R_3 = 3R_f$$

Substitute $50\,\mathrm{k\Omega}$ for R_f.

$$R_3 = 3\left(50 \times 10^3\right)$$
$$R_3 = 150\,\mathrm{k\Omega}$$

Therefore, the value of R_3, is $\boxed{150\,\mathrm{k\Omega}}$.